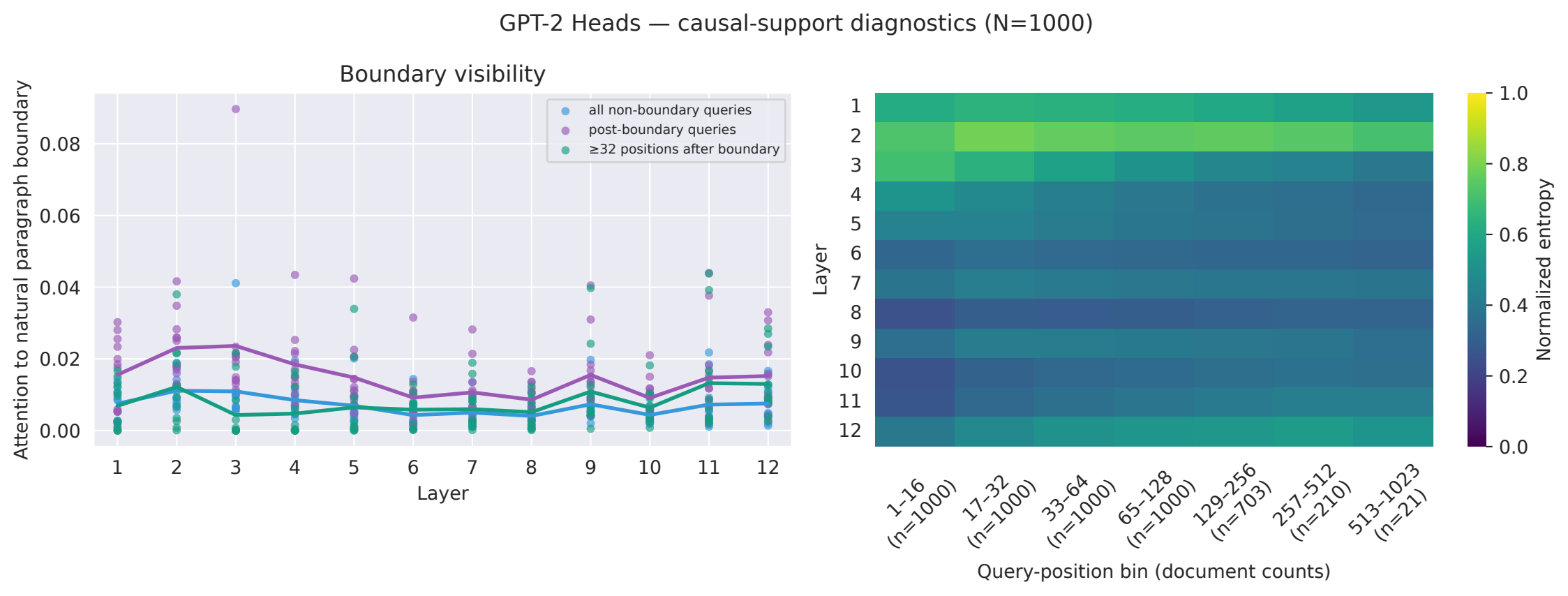

GPT-2 Heads — causal-support diagnostics (N=1000)
Boundary visibility
Attention to natural paragraph boundary
0.00
0.02
0.04
0.06
0.08
all non-boundary queries
post-boundary queries
≥32 positions after boundary
1
2
3
4
5
6
7
8
9
10
11
12
Layer
Layer
1
2
3
4
5
6
7
8
9
10
11
12
1–16 (n=1000)
17–32 (n=1000)
33–64 (n=1000)
65–128 (n=1000)
129–256 (n=703)
257–512 (n=210)
513–1023 (n=21)
Query-position bin (document counts)
Normalized entropy
0.0
0.2
0.4
0.6
0.8
1.0